

- 13 A particle of a mass of 90.0 g undergoes simple harmonic motion. The graph in Fig. 13 shows the variation of its kinetic energy, E_K with time, t .

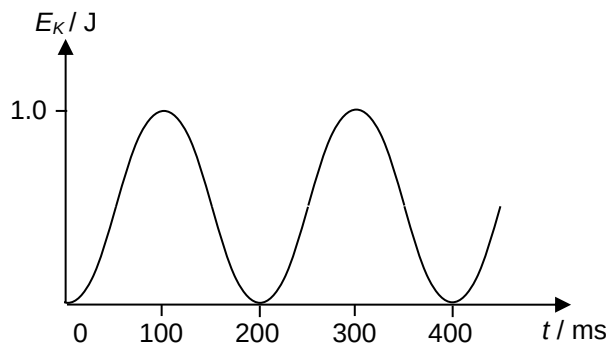


Fig. 13

What is the maximum acceleration of the particle?

- A 0.074 m s^{-2} B 0.148 m s^{-2} C 37 m s^{-2} D 74 m s^{-2}

Ans: D

$$T = 400\text{ms}$$

$$\max ke = \frac{1}{2}m\omega^2x_o^2 = 1$$

$$x_o = 0.3\text{m}$$

$$a_{\max} = 74\text{ms}^{-2}$$

- 14 A conical pendulum undergoes uniform circular motion in a horizontal plane. The radius of the